# IKON SKILLS™

*ikonskills.ac*

# ASSESSMENT ENGINE

Developer Specification Document

*Version 3.0 | Pay-Per-MC Architecture | 14 AI Interaction Points Per Core Competency | Claude Sonnet Throughout*

| | |
|---|---|
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| Role | Founder & Director General, EIU-Paris \| Founder & CEO, IKON Educational and Psychological Consultancy |
| Platform | ikonskills.ac \| Launching June 2026 |
| AI Model | Claude Sonnet (Anthropic) — all AI Interaction Points |
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# 1. Purpose and Scope

**This document is the definitive technical and pedagogical specification for the IKON SKILLS™ assessment engine at ikonskills.ac. It is addressed to the development team and covers four binding requirements:**

1. **One:** The binary competency model: the only two outcomes are COMPETENT or NOT YET COMPETENT, determined by a 75% pass threshold.
2. **Two:** The 14 AI Interaction Points (AIPs) architecture: the exact ceiling on live Claude Sonnet API calls per Core Competency, which controls platform operating costs with precision.
3. **Three:** The question bank pre-generation pipeline: how reattempt question variants are built at MC publish time, not at practitioner learning time, so reattempts cost zero additional API calls.
4. **Four:** The Pay-Per-MC pricing model at $14.99 per Micro-Credential: the sole commercial model for the platform at launch, and the unit economics that the AIP architecture must protect.

**DEVELOPER PRIORITY: Read Sections 3, 4, and 5 before writing any code that touches the AI API. Every AI API call on this platform must map to a named AIP. Unmapped API calls are a cost and architecture violation.**

## 1.1 Terminology Reference

| Term | Definition |
|---|---|
| Micro-Credential (MC) | A standalone credential within a Category. Contains exactly 10 Core Competencies. Priced at $14.99 per MC. |
| Core Competency (CC) | One of 10 learning units within an MC. Each CC has its own content, 3 Formative Assessments, and 1 Summative Assessment. Allocated exactly 14 AI Interaction Points. |
| AI Interaction Point (AIP) | A single live call to the Claude Sonnet API. The platform has exactly 14 named AIPs per CC. No unnamed or unmapped API calls are permitted. |
| Question Bank | A pre-generated library of assessment question variants built at MC publish time via a separate Sonnet pipeline. Served during reattempts at zero additional API cost. |
| IKON Practitioner | The person learning on the platform. Never 'student', 'learner', or 'user' in any UI text. |
| Statement of Competency | The official IKON record of competencies achieved. Never called a transcript. |
| Developing Competency | The remedial learning phase triggered when an assessment is not passed. Pre-authored static content. Never AI-generated on demand. Not an outcome label. |
| Clean Run | An MC journey in which the IKON Practitioner passes every assessment on the first attempt. Costs exactly 141 live API calls (14 per CC x 10 CCs + 1 AIP-14). |

## 1.2 Platform Scale

| Category | MCs | CCs | AIPs (clean run) |
|---|---|---|---|
| Cat 01: AI, Automation & Digital Intelligence | 15 | 150 | 2,115 |
| Cat 02: Data, Analytics & Insight | 15 | 150 | 2,115 |
| Cat 03: Business, Strategy & Leadership | 15 | 150 | 2,115 |
| Cat 04: Project, Product & Operations | 15 | 150 | 2,115 |
| Cat 05: Technology & Digital Enablement | 10 | 100 | 1,410 |
| Cat 06: Marketing, Sales & Growth | 10 | 100 | 1,410 |
| Cat 07: HR, Learning & People Development | 10 | 100 | 1,410 |
| Cat 08: Finance, Compliance & ESG | 10 | 100 | 1,410 |
| Cat 09: Teaching & Education (BEd-MEd-EdD) | 48 | 480 | 6,768 |
| Cat 10: Brand Strategy & Leadership (TWBF) | 36 | 360 | 5,076 |
| **TOTAL** | **184** | **1,840** | **25,944 AIPs** |

**Note: Per-MC AIP count = (14 x 10 CCs) + 1 AIP-14 = 141. Total platform AIPs = 184 MCs x 141 = 25,944 live Sonnet calls for one cohort on a clean run.**

# 2. The Binary Competency Model

**IKON SKILLS™ operates on a strict binary competency model. Every assessment event at every level produces exactly one of two outcomes. There are no grades, percentages shown as final results, or intermediate states.**

| | |
|---|---|
| **COMPETENT** | The IKON Practitioner scored >= 75% on the required assessment. This CC assessment gate is cleared. The next gate or CC unlocks. The score is stored internally but is never the primary display. |
| **NOT YET COMPETENT** | The IKON Practitioner scored < 75% on the required assessment. The gate remains locked. The Developing Competency phase is triggered. The IKON Practitioner reviews pre-authored remedial content and reattempts using a pre-generated question variant from the Question Bank. |

**NEVER build, display, or store any outcome state other than COMPETENT or NOT YET COMPETENT as a final result. Labels such as 'Partially Competent', 'In Progress', 'Borderline', 'Developing', or any numeric score as a primary result are prohibited in all UI-facing text.**

## 2.1 The 75% Threshold: Exact Implementation

| | |
|---|---|
| **Pass Formula** | score >= 0.75 * total_marks |
| **Rounding Rule** | Round to 2 decimal places. 74.99% = NOT YET COMPETENT. 75.00% = COMPETENT. No rounding up. |
| **Independence** | Each FA and the SA are assessed independently against 75%. No averaging of scores across assessments. Passing FA1 and FA3 while failing FA2 means NOT YET COMPETENT at FA2. |
| **Score Storage** | All scores stored internally with timestamp and attempt number. Scores are never the primary display to the IKON Practitioner. |
| **Score Display** | The system may display the score as secondary information after showing the binary outcome label. Example: 'Not Yet Competent (68%)'. The label always comes first. |

# 3. The 14 AI Interaction Points: Complete Architecture

**Every Core Competency is allocated exactly 14 AI Interaction Points. An AIP is a single live call to the Claude Sonnet API. The 14 AIPs are the total budget for delivering, assessing, and confirming one CC for one IKON Practitioner. This budget is fixed, named, and non-negotiable.**

**DEVELOPER RULE: Every call to the Claude Sonnet API must map to one of the 14 named AIPs below. If you are writing code that calls the API and cannot identify which AIP it maps to, stop and escalate. Unmapped API calls are a cost violation and an architecture defect.**

## 3.1 The 14 AIPs: Full Reference Table

| AIP | Name | Type | Fires When | What Claude Does |
|---|---|---|---|---|
| **AIP-01** | **Concept Delivery** | **Teach** | *CC begins, first access only* | Delivers the core concepts, definitions, and foundational knowledge of this CC. Tailored to the CC subject. Cached after first delivery. Never called again for this practitioner in this CC. |
| **AIP-02** | **Worked Example** | **Teach** | *After AIP-01 completes* | Presents a detailed worked example or scenario walkthrough demonstrating the CC concepts in an applied real-world context. This is what separates IKON from a standard LMS. Cached after first delivery. |
| **AIP-03** | **Reflective Prompt** | **Teach** | *After AIP-02 completes* | Poses a structured reflective question to the IKON Practitioner and responds to their input with personalised guidance. Deepens engagement before assessment begins. Cached after first exchange. |
| **AIP-04** | **Pre-Assessment Consolidation** | **Teach** | *After AIP-03 completes* | Delivers a concise synthesis of the CC: key takeaways, common misconceptions, and what the upcoming assessments will test. Frames the practitioner's mindset for FA1. Cached after first delivery. |
| **AIP-05** | **FA1 Generation** | **Assess** | *After AIP-04 completes* | Generates the Formative Assessment 1 question set (recall and comprehension of content just taught). Live call on first attempt only. Reattempts use pre-generated Question Bank variants. |
| **AIP-06** | **FA1 Feedback** | **Remediate** | *After FA1 is submitted* | Delivers outcome feedback. COMPETENT: brief affirmation and bridge to FA2. NOT YET COMPETENT: targeted thematic guidance and instruction to review Developing Competency content. Cached after first failure feedback. Reattempt failures re-serve cached response. |

| | | | | |
|---|---|---|---|---|
| **AIP-07** | **FA2 Generation** | **Assess** | *After FA1 = COMPETENT* | Generates the Formative Assessment 2 question set. Varies in question style from FA1 to provide genuine reinforcement variety. Live call on first attempt. Reattempts use Question Bank variants. |
| **AIP-08** | **FA2 Feedback** | **Remediate** | *After FA2 is submitted* | Delivers outcome feedback for FA2. Same caching logic as AIP-06. COMPETENT bridges to FA3. NOT YET COMPETENT triggers Developing Competency phase. |
| **AIP-09** | **FA3 Generation** | **Assess** | *After FA2 = COMPETENT* | Generates the Formative Assessment 3 question set. Final reinforcement gate before the Summative Assessment. Live call on first attempt. Reattempts use Question Bank variants. |
| **AIP-10** | **FA3 Feedback** | **Remediate** | *After FA3 is submitted* | Delivers outcome feedback for FA3. Same caching logic as AIP-06. COMPETENT unlocks the Summative Assessment. NOT YET COMPETENT triggers Developing Competency phase. |
| **AIP-11** | **SA Generation** | **Assess** | *After FA1 + FA2 + FA3 = COMPETENT* | Generates the Summative Assessment: an applied, scenario-based or performance-based assessment requiring demonstration of CC competency in context. This is never a recall question. Live call on first attempt. Reattempts use Question Bank variants. |
| **AIP-12** | **SA Feedback** | **Remediate** | *After SA is submitted* | Delivers outcome feedback for the SA. COMPETENT: flags CC as COMPETENT, triggers AIP-13. NOT YET COMPETENT: delivers targeted Developing Competency guidance. Cached after first failure. Reattempt failures re-serve cached response. |
| **AIP-13** | **CC Competency Summary** | **Confirm** | *When CC is flagged COMPETENT (SA passed)* | Delivers a structured summary of what the IKON Practitioner demonstrated across this CC. Affirms the competency achieved. Updates the Statement of Competency. Unlocks the next CC. Fires only on CC pass. Never fires on a failed attempt. |
| **AIP-14** | **MC Completion Reflection** | **Confirm** | *When ALL 10 CCs = COMPETENT* | Fires exactly once per MC per practitioner when all 10 CCs are flagged COMPETENT. Delivers a closing synthesis across the full MC journey. Bridges to badge and Certificate of Completion issuance. Never fires on partial completion. |

## 3.2 AIP Distribution by Type

| Type | Count | AIPs | Purpose |
|---|---|---|---|
| Teach | 4 | AIP-01 to AIP-04 | Content delivery, worked example, reflection, consolidation. The genuine learning experience before any assessment. |
| Assess | 3 | AIP-05, AIP-07, AIP-09, AIP-11 | FA1, FA2, FA3, and SA generation. Live on first attempt. Reattempts use Question Bank. |
| Remediate | 4 | AIP-06, AIP-08, AIP-10, AIP-12 | Outcome feedback for each assessment. Cached after first failure. Re-served on subsequent failures at zero API cost. |
| Confirm | 2 | AIP-13, AIP-14 | CC competency summary (on CC pass) and MC completion reflection (on all-10-CC pass). Conditional AIPs. |
| **TOTAL** | **13 per CC + 1 per MC** | **AIP-01 to AIP-14** | **14 AIPs named. AIP-14 fires once per MC not once per CC. Clean-run cost: 141 Sonnet calls per MC.** |

**IMPORTANT: AIP-05, AIP-07, AIP-09, AIP-11 are listed as 4 Assess AIPs but the Type row says 3. Correction for developers: there are 4 assessment generation AIPs (FA1, FA2, FA3, SA) and 4 remediation AIPs (one feedback per assessment). Total = 4 Teach + 4 Assess + 4 Remediate + 2 Confirm = 14. This is correct.**

# 4. Assessment Sequence: The Complete Learning Path

**Every Core Competency follows the same fixed sequence. The sequence is enforced programmatically. No step may be skipped, reordered, or bypassed. The table below shows the clean-run path (all assessments passed on first attempt) with the AIP that fires at each step.**

## 4.1 Clean Run Sequence

| Step | AIP | Phase | System Action |
|---|---|---|---|
| 1 | AIP-01 | Concept Delivery | Claude delivers CC concept content. Response cached. IKON Practitioner reads and engages. |
| 2 | AIP-02 | Worked Example | Claude presents a worked real-world example or scenario for this CC. Response cached. |
| 3 | AIP-03 | Reflective Prompt | Claude poses a reflective question. IKON Practitioner responds. Claude replies with personalised guidance. Exchange cached. |
| 4 | AIP-04 | Pre-Assessment Consolidation | Claude delivers a synthesis of key CC takeaways and frames what FA1 will assess. Response cached. |
| 5 | AIP-05 | FA1 Generation | Claude generates FA1 question set (recall and comprehension). Live API call. IKON Practitioner attempts FA1. |
| 6 | AIP-06 | FA1 Feedback | Score >= 75%: Claude delivers brief affirmation and bridges to FA2. FA2 unlocks. |
| 7 | AIP-07 | FA2 Generation | Claude generates FA2 question set (different style from FA1). Live API call. IKON Practitioner attempts FA2. |
| 8 | AIP-08 | FA2 Feedback | Score >= 75%: Claude delivers affirmation and bridges to FA3. FA3 unlocks. |
| 9 | AIP-09 | FA3 Generation | Claude generates FA3 question set (final reinforcement gate). Live API call. IKON Practitioner attempts FA3. |
| 10 | AIP-10 | FA3 Feedback | Score >= 75%: Claude delivers affirmation and bridges to Summative Assessment. SA unlocks. |
| 11 | AIP-11 | SA Generation | Claude generates the applied Summative Assessment scenario. Live API call. IKON Practitioner attempts SA. |
| 12 | AIP-12 | SA Feedback | Score >= 75%: Claude confirms competency. CC flagged COMPETENT. Triggers AIP-13. |
| 13 | AIP-13 | CC Competency Summary | Claude delivers a structured summary of what was demonstrated in this CC. Statement of Competency updated. Next CC unlocks. |
| 14 | AIP-14 | MC Completion Reflection | Fires ONCE when all 10 CCs = COMPETENT. Claude delivers MC closing synthesis. Badge and Certificate of Completion issued. |

## 4.2 The Failure Path: What Happens When an Assessment Is Not Passed

**When an IKON Practitioner scores below 75% on any assessment (FA1, FA2, FA3, or SA), the following sequence fires:**

5. **Step 1:** The feedback AIP for that assessment fires immediately as a live Claude Sonnet call. It delivers targeted thematic guidance (not answer reveals). This response is stored as the cached failure feedback for this assessment.
6. **Step 2:** The Developing Competency phase is triggered. Pre-authored static remedial content is served from the platform content library. This is NOT a live API call. It is authored content built into the MC at publish time.
7. **Step 3:** The IKON Practitioner reviews the Developing Competency content and signals readiness to reattempt.
8. **Step 4:** The system serves a pre-generated question variant from the Question Bank for this assessment. This is NOT a live API call. It is drawn from the 3 variants pre-generated at MC publish time.
9. **Step 5:** The IKON Practitioner attempts the reattempt. The system scores it against the 75% threshold.
10. **Step 6a:** If COMPETENT: the feedback AIP fires as a live call (pass feedback is always live, not cached). The gate clears and the next step unlocks.
11. **Step 6b:** If NOT YET COMPETENT again: the system re-serves the cached failure feedback (no new API call). A fresh Question Bank variant is served for the next reattempt. This loop continues until the assessment is passed.

**THE COST RULE IN ONE SENTENCE: The failure feedback AIP fires as a live call exactly once per assessment per practitioner per CC. Every subsequent failure for the same assessment re-serves the cached response. Zero additional API cost per reattempt after the first failure.**

## 4.3 Formative vs Summative: The Design Distinction

**This table exists because developers must build two separate question generation pathways with different prompt architectures in the AI API calls.**

| | Formative Assessment (FA1, FA2, FA3) | Summative Assessment (SA) |
|---|---|---|
| Purpose | Reinforce and confirm recall of content just taught in AIP-01 through AIP-04 | Demonstrate applied competency in a realistic professional context |
| Question Style | Knowledge check: definitions, concepts, matching, recall, short comprehension | Scenario, case study, applied problem, or professional simulation |
| AIP | AIP-05, AIP-07, AIP-09 | AIP-11 |
| Count per CC | 3 (FA1, FA2, FA3 in sequence) | 1 |
| Gate Function | Must pass to unlock next FA or SA | Must pass to flag CC = COMPETENT |
| Reattempt Source | Question Bank variant (pre-generated at MC publish time) | Question Bank variant (pre-generated at MC publish time) |
| Feedback AIP | AIP-06, AIP-08, AIP-10 | AIP-12 |
| Feedback Caching | First failure: live call. Subsequent failures: cached response. | First failure: live call. Subsequent failures: cached response. |

# 5. The Question Bank Pre-Generation Pipeline

**The Question Bank is the mechanism that makes unlimited reattempts economically viable. It is a library of pre-generated assessment question variants built by a separate Claude Sonnet pipeline when each MC is published. It is never generated on demand during a live practitioner session.**

## 5.1 The Question Bank Math

| | |
|---|---|
| **Assessments per CC** | 4 (FA1, FA2, FA3, SA) |
| **Variants per assessment** | 3 (Variant A, Variant B, Variant C) |
| **Question sets per CC** | 4 assessments x 3 variants = 12 pre-generated question sets |
| **CCs per MC** | 10 |
| **Total question sets per MC** | 12 x 10 = 120 pre-generated question sets per MC |
| **When generated** | Once, at MC publish time, via a dedicated Sonnet pipeline job. Never during a practitioner session. |
| **API calls for pipeline** | 120 Sonnet calls per MC at publish time. This is a one-time build cost per MC, not a per-practitioner cost. |

## 5.2 How Variants Are Served During Reattempts

**The system must follow a defined rotation logic when serving Question Bank variants. Here are the rules:**

12. **Rule 1:** First attempt at any assessment: the system calls AIP-05, AIP-07, AIP-09, or AIP-11 as a live Sonnet call to generate questions fresh for this practitioner session.
13. **Rule 2:** First reattempt: serve Variant A from the Question Bank. No live API call.
14. **Rule 3:** Second reattempt: serve Variant B from the Question Bank. No live API call.
15. **Rule 4:** Third reattempt and beyond: serve Variant C from the Question Bank, then cycle back to Variant A. No live API call at any point.
16. **Rule 5:** The practitioner never sees the same variant twice in consecutive attempts. Rotation is: Live, A, B, C, A, B, C...

**DEVELOPER INSTRUCTION: The Question Bank must be stored in the database at the MC level, not the practitioner level. All practitioners attempting the same CC draw from the same pre-generated variant pool. This is not personalised per-practitioner at the reattempt stage. It is pooled content served in rotation.**

## 5.3 Question Bank Build Pipeline: Developer Specification

**The pipeline is a separate server-side job triggered when an MC is published. It must not run during practitioner sessions.**

| Pipeline Parameter | Specification |
|---|---|
| Trigger | MC status transitions to PUBLISHED. Pipeline runs once. If MC content is edited and republished, pipeline reruns and overwrites existing variants. |
| Model | Claude Sonnet. Same model as live AIPs for consistency of question quality. |
| Prompt input | CC name, CC competency descriptors, the 10 Core Competency titles for this MC, and the question style specification for each assessment type (recall for FAs, applied for SA). |
| Output per call | One complete question set per variant per assessment: questions, answer options (for MCQ), correct answer key, and difficulty calibration note. |
| Storage | question_bank table: mc_id, cc_id, assessment_type (FA1\|FA2\|FA3\|SA), variant (A\|B\|C), question_set (JSON), generated_at (TIMESTAMP). |
| Error handling | If any variant generation fails, retry up to 3 times before flagging the MC as PUBLISH_FAILED. An MC with incomplete Question Bank variants must not be made available to practitioners. |
| Cost | 120 Sonnet calls per MC at publish time. At $0.0135 per call: $1.62 per MC as a one-time build cost. Amortised across all practitioners who purchase that MC, this approaches zero per practitioner at any meaningful volume. |
| Separation | This pipeline runs on a separate worker process or serverless function. It must not share infrastructure with the live practitioner session API call handler. A slow build job must never delay a practitioner's learning session. |

# 6. Caching Rules: What Is Stored and When

**Caching is the second mechanism (alongside the Question Bank) that keeps the live API cost fixed at 14 AIPs per CC per practitioner regardless of how many reattempts occur. These caching rules are mandatory. They are not optimisations. They are cost-control requirements.**

## 6.1 What Must Be Cached Per Practitioner Per CC

| AIP | Cache Trigger | Cache Rule |
|---|---|---|
| AIP-01 to AIP-04 | First delivery | Store the full Claude response. On any subsequent access (practitioner revisits the CC), serve from cache. Never call the API again for teaching content within the same CC for the same practitioner. |
| AIP-03 | First reflective exchange | Cache the practitioner's input and Claude's response. On revisit, display the cached exchange. Do not re-prompt Claude. |
| AIP-06 | First FA1 failure | Cache the NOT YET COMPETENT feedback response. Serve from cache on all subsequent FA1 failures. Fire a new live call only when FA1 is passed (COMPETENT feedback is always live). |
| AIP-08 | First FA2 failure | Same rule as AIP-06. Cache failure feedback. Live call on pass only. |
| AIP-10 | First FA3 failure | Same rule as AIP-06. Cache failure feedback. Live call on pass only. |
| AIP-12 | First SA failure | Same rule as AIP-06. Cache failure feedback. Live call on pass only. |
| AIP-13 | CC completion | Cache the CC Competency Summary. Display on dashboard and Statement of Competency. Never regenerate. |
| AIP-14 | MC completion | Cache the MC Completion Reflection. Display on completion dashboard. Never regenerate. aip14_fired flag = TRUE prevents re-firing. |

## 6.2 What Must Never Be Cached

- COMPETENT outcome feedback from any AIP (AIP-06, AIP-08, AIP-10, AIP-12): always a live call. The pass moment must feel fresh and personalised.
- AIP-11 SA generation on the first attempt: always a live call. The applied scenario is generated fresh for this practitioner on their first SA attempt.
- AIP-05, AIP-07, AIP-09 on the first attempt: always a live call.

## 6.3 The Caching Decision Tree (For Every AIP Call)

**Before making any API call, the system must check this decision tree in order:**

17. Is this a Teaching AIP (AIP-01 to AIP-04)? If yes: is there a cached response for this practitioner and CC? If yes: serve cache. If no: call API and cache response.
18. Is this a Generation AIP (AIP-05, AIP-07, AIP-09, AIP-11)? If yes: is this the first attempt? If yes: call API. If no: serve Question Bank variant. Do not call API.
19. Is this a Remediate AIP (AIP-06, AIP-08, AIP-10, AIP-12)? If yes: is the outcome COMPETENT? If yes: call API (pass feedback is always live). If no: is there a cached failure response? If yes: serve cache. If no: call API and cache response.
20. Is this AIP-13? If yes: call API (fires only on CC pass, never duplicates). Cache response.
21. Is this AIP-14? If yes: has aip14_fired already been set to TRUE for this practitioner and MC? If yes: do not call API. If no: call API, cache response, set aip14_fired = TRUE.

# 7. Micro-Credential Completion Logic

**A Micro-Credential is completed when all 10 Core Competencies are individually flagged COMPETENT. There is no alternative path, partial credential, or workaround.**

## 7.1 MC Completion Rules

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|---|---|
| **Completion Condition** | COUNT(cc_status = COMPETENT) = 10 for this practitioner and MC. All 10. Not 9. Not 8. |
| **Partial Completion** | No partial MC credential exists. 9 of 10 COMPETENT = MC is not complete. No badge or certificate issued. No AIP-14 fired. |
| **Digital Badge** | Issued automatically when mc_status transitions to COMPLETED. Flat-top hexagon format per IKON SKILLS™ brand specification. |
| **Certificate of Completion** | Issued alongside the badge on MC completion. References EIU-Paris quality assurance. Template-driven. Not AI-generated. |
| **AIP-14** | Fires exactly once when mc_status = COMPLETED. Sets aip14_fired = TRUE. Does not fire again even if practitioner revisits the MC. |
| **Statement of Competency** | Updated in real time as each CC is flagged COMPETENT. Records CC name, MC name, and date of competency achievement. Database-driven. Not AI-generated. |
| **CC Sequencing** | CC-02 unlocks only when CC-01 = COMPETENT. CC-03 unlocks only when CC-02 = COMPETENT. And so on through CC-10. CCs are not available in parallel. |

# 8. Pricing Model and Unit Economics

**IKON SKILLS™ launches with a single pricing model: Pay-Per-MC at $14.99 per Micro-Credential. There are no subscription tiers at launch. This section documents the unit economics so the development team understands the cost boundaries the AIP architecture is designed to protect.**

## 8.1 Pay-Per-MC: The Only Pricing Model at Launch

| | |
|---|---|
| **Price** | $14.99 per Micro-Credential |
| **Includes** | Full AI-powered learning journey across all 10 CCs. Unlimited competency reattempts until the 75% benchmark is achieved. Digital badge and Certificate of Completion on MC completion. Access to Statement of Competency dashboard. |
| **Reattempts** | Unlimited. Reattempts draw from the pre-generated Question Bank at zero additional API cost to the platform. |
| **Correct Public Language** | "Each Micro-Credential includes a full AI-powered learning journey with unlimited competency reattempts until the 75% benchmark is achieved." |
| **Do Not Say** | "One assessment attempt per MC." This contradicts the competency model and the unlimited reattempt architecture. Remove this language from all public-facing materials. |

## 8.2 Unit Economics Per MC Sold

| Line Item | Amount | Notes |
|---|---|---|
| Gross Revenue per MC | $14.99 | |
| Payment Processing (Stripe 2.9% + $0.30) | -$0.73 | Approximate. Varies by card type and region. |
| **Net Revenue** | **$14.26** | |
| AI Cost: 141 live Sonnet calls | -$1.90 | 141 AIPs x $0.0135 per call. Clean run. |
| Question Bank Build (amortised) | -$0.02 | $1.62 build cost per MC / estimated 100+ purchasers |
| White-Label Partner Share (20%) | -$3.00 | Where applicable. 20% of $14.99. |
| **Net Margin (with partner share)** | **$9.34** | **62% gross margin. Healthy at launch scale.** |
| **Net Margin (direct sale, no partner)** | **$12.34** | **82% gross margin. Direct sales or own traffic.** |

## 8.3 Reattempt Cost Sensitivity

**Because reattempts use the Question Bank (zero API cost) and cached feedback (zero API cost), the AI cost is not sensitive to reattempt frequency. The only additional live API cost from reattempts is the COMPETENT feedback call when a practitioner eventually passes an assessment they previously failed.**

| | |
|---|---|
| **Maximum realistic additional cost** | If a practitioner fails all 4 assessments once each and then passes: 4 extra COMPETENT feedback calls = 4 x $0.0135 = $0.054 additional AI cost per MC. Margin impact: negligible. |
| **Worst case scenario** | A practitioner who fails every assessment 10 times across 10 CCs still only adds: 40 extra COMPETENT feedback calls = $0.54 additional AI cost. Total AI cost: $2.44. Margin remains strong at $11.82 per MC direct sale. |
| **Conclusion** | The 14-AIP architecture with Question Bank and caching makes the margin robust against even extreme reattempt behaviour. This is by design. |

# 9. Database and State Management

**The following is the minimum required data model. Every field listed is necessary for the assessment engine, caching system, and Question Bank to function correctly.**

## 9.1 Per Core Competency, Per IKON Practitioner

| Field | Type | Notes |
|---|---|---|
| practitioner_id | UUID | Unique IKON Practitioner identifier |
| mc_id | UUID | Micro-Credential identifier |
| cc_id | UUID | Core Competency identifier |
| cc_status | ENUM: COMPETENT \| NOT_YET_COMPETENT \| IN_PROGRESS | Current CC status |
| aip01_cache | TEXT \| NULL | Cached AIP-01 Concept Delivery response |
| aip02_cache | TEXT \| NULL | Cached AIP-02 Worked Example response |
| aip03_cache | TEXT \| NULL | Cached AIP-03 Reflective Prompt exchange |
| aip04_cache | TEXT \| NULL | Cached AIP-04 Pre-Assessment Consolidation response |

| | | |
|---|---|---|
| fa1_status | ENUM: PASSED \| NOT_YET_PASSED \| NOT_STARTED | FA1 gate status |
| fa1_attempts | INTEGER | Total FA1 attempts including reattempts |
| fa1_last_score | DECIMAL (0.00-100.00) | Internal only. Not displayed as primary result. |
| fa1_feedback_fail_cache | TEXT \| NULL | Cached AIP-06 failure feedback. Served on 2nd+ failures. |
| fa1_variant_index | INTEGER | Current Question Bank variant rotation index for FA1 |
| fa2_status | ENUM: PASSED \| NOT_YET_PASSED \| NOT_STARTED \| LOCKED | LOCKED until fa1_status = PASSED |
| fa2_attempts | INTEGER | Total FA2 attempts |
| fa2_last_score | DECIMAL (0.00-100.00) | Internal only |
| fa2_feedback_fail_cache | TEXT \| NULL | Cached AIP-08 failure feedback |
| fa2_variant_index | INTEGER | Question Bank variant rotation index for FA2 |
| fa3_status | ENUM: PASSED \| NOT_YET_PASSED \| NOT_STARTED \| LOCKED | LOCKED until fa2_status = PASSED |
| fa3_attempts | INTEGER | Total FA3 attempts |
| fa3_last_score | DECIMAL (0.00-100.00) | Internal only |
| fa3_feedback_fail_cache | TEXT \| NULL | Cached AIP-10 failure feedback |
| fa3_variant_index | INTEGER | Question Bank variant rotation index for FA3 |
| sa_status | ENUM: PASSED \| NOT_YET_PASSED \| NOT_STARTED \| LOCKED | LOCKED until fa1 + fa2 + fa3 = PASSED |
| sa_attempts | INTEGER | Total SA attempts |
| sa_last_score | DECIMAL (0.00-100.00) | Internal only |
| sa_feedback_fail_cache | TEXT \| NULL | Cached AIP-12 failure feedback |
| sa_variant_index | INTEGER | Question Bank variant rotation index for SA |
| aip13_cache | TEXT \| NULL | Cached AIP-13 CC Competency Summary. Set on CC pass. |

| | | |
|---|---|---|
| cc_competent_date | TIMESTAMP \| NULL | Date CC was flagged COMPETENT. NULL until achieved. |
| live_api_calls_used | INTEGER | Running count of live Sonnet calls for this CC. Alert if > 14 on clean run. |

## 9.2 Per Micro-Credential, Per IKON Practitioner

| Field | Type | Notes |
|---|---|---|
| mc_status | ENUM: IN_PROGRESS \| COMPLETED | COMPLETED only when all 10 cc_status = COMPETENT |
| mc_completion_date | TIMESTAMP \| NULL | Set when mc_status = COMPLETED |
| badge_issued | BOOLEAN | TRUE only when mc_status = COMPLETED |
| certificate_issued | BOOLEAN | TRUE only when mc_status = COMPLETED |
| aip14_cache | TEXT \| NULL | Cached AIP-14 MC Completion Reflection |
| aip14_fired | BOOLEAN | TRUE after AIP-14 fires. Prevents duplicate calls. |
| total_live_api_calls | INTEGER | Running total across all CCs. Alert threshold: > 141 on clean run. |
| purchase_amount | DECIMAL | $14.99 at time of purchase. Stored for audit. |
| purchased_at | TIMESTAMP | Date and time of MC purchase |

## 9.3 Question Bank Table

| Field | Type | Notes |
|---|---|---|
| mc_id | UUID | Micro-Credential this bank belongs to |
| cc_id | UUID | Core Competency this bank belongs to |
| assessment_type | ENUM: FA1 \| FA2 \| FA3 \| SA | Which assessment this variant serves |
| variant | ENUM: A \| B \| C | Variant identifier. Three per assessment per CC. |
| question_set | JSON | Full question set: questions, options, answer key, difficulty notes |
| generated_at | TIMESTAMP | When the pipeline generated this variant |
| pipeline_version | VARCHAR | Version of the build pipeline that generated this variant. Useful for quality audits. |

# 10. Language Standards for All UI Text

**These terminology rules apply to every text string in the platform UI, email notifications, dashboard labels, and any AI prompt that includes user-facing output. They are institutional standards set by EIU-Paris and IKON, not style suggestions.**

| NEVER Use | ALWAYS Use |
|---|---|
| Failed / Incorrect / Wrong | Not Yet Competent |
| Try again | Review and reattempt |
| Student / Learner / User | IKON Practitioner |
| Transcript | Statement of Competency |
| Courses | Micro-Credentials |
| Lessons | Core Competencies |
| Test / Exam / Quiz | Assessment |
| Passed | Competent |
| Developing Competency (as a result label) | Not Yet Competent |
| One assessment attempt per MC | Full AI-powered learning journey with unlimited competency reattempts |
| Score shown as primary result | Binary outcome label first: COMPETENT or NOT YET COMPETENT |
| Subscription / Plan | Not applicable at launch. Pay-Per-MC only. |

# 11. Quick Reference: 18 Binding Rubric Rules

**Every rule below is binding. No exceptions, workarounds, or deviations may be implemented without written approval from Hon. Ky. Col. Prof. Dr. Edward Roy Krishnan.**

| # | Rule | Specification |
|---|---|---|
| **R-01** | **Binary Outcome Only** | Exactly two outcomes: COMPETENT (>= 75%) or NOT YET COMPETENT (< 75%). No partial, borderline, or intermediate states in any UI or database field. |
| **R-02** | **75% Threshold** | 75.00% is the minimum passing score for every formative and summative assessment. 74.99% is NOT YET COMPETENT. No rounding up. |
| **R-03** | **4 Teaching AIPs per CC** | Every CC must receive AIP-01 (Concept Delivery), AIP-02 (Worked Example), AIP-03 (Reflective Prompt), and AIP-04 (Pre-Assessment Consolidation) before any assessment is presented. |
| **R-04** | **3 Formative Assessments per CC** | Exactly FA1, FA2, FA3 in sequence. Each must be independently passed at 75% before the next unlocks. Cannot be skipped or merged. |
| **R-05** | **1 Summative Assessment per CC** | Exactly 1 SA per CC. Applied and scenario-based. Unlocks only after FA1 + FA2 + FA3 are all COMPETENT. |
| **R-06** | **Sequential Gating** | FA1 passed before FA2. FA2 passed before FA3. FA3 passed before SA. SA passed before CC is COMPETENT. Enforced programmatically. No bypass. |
| **R-07** | **CC Flag Condition** | CC = COMPETENT only after SA is passed at >= 75%. Passing all 3 FAs alone does not clear a CC. |
| **R-08** | **MC Completion Gate** | MC = COMPLETED only when all 10 CCs = COMPETENT. No partial MC credential. |
| **R-09** | **Badge and Certificate Gate** | Digital badge and Certificate of Completion issued only when mc_status = COMPLETED. |
| **R-10** | **14 AIPs Hard Ceiling** | Each CC is allocated exactly 14 named AIPs. No unmapped API calls. Alert engineering if live_api_calls_used exceeds 14 on a clean run. |
| **R-11** | **Teaching AIPs Cached** | AIP-01 through AIP-04 fire as live calls once per practitioner per CC. All subsequent accesses serve cached responses. Never re-call the API for teaching content. |
| **R-12** | **Failure Feedback Cached** | AIP-06, AIP-08, AIP-10, AIP-12 fire as live calls once on the first failure. All subsequent failures for the same assessment serve the cached response. Zero additional API cost per reattempt after the first failure. |
| **R-13** | **Pass Feedback Always Live** | COMPETENT outcome feedback from AIP-06, AIP-08, AIP-10, AIP-12 is always a live Sonnet call. The pass moment is never served from cache. |

| | | |
|---|---|---|
| **R-14** | **Reattempts Use Question Bank** | From the second attempt onward, all assessment questions are served from the pre-generated Question Bank. No live API call for question generation on reattempts. |
| **R-15** | **Question Bank Built at Publish** | 120 question sets per MC (3 variants x 4 assessments x 10 CCs) are generated by a separate pipeline at MC publish time. Never generated during a practitioner session. |
| **R-16** | **AIP-13 Is Conditional** | AIP-13 (CC Competency Summary) fires only when CC is flagged COMPETENT. Never on a failed SA attempt. |
| **R-17** | **AIP-14 Fires Once Per MC** | AIP-14 (MC Completion Reflection) fires exactly once per practitioner per MC. aip14_fired = TRUE prevents re-firing. |
| **R-18** | **Pay-Per-MC Only** | $14.99 per MC. No subscription tiers at launch. Every MC purchase includes unlimited reattempts via Question Bank. The platform must not display or offer subscription pricing. |

# 12. Authorization and Escalation

**This document is the definitive specification for the IKON SKILLS™ assessment engine. Any deviation from any rule in this document requires written approval from:**


## Hon. Ky. Col. Prof. Dr. Edward Roy Krishnan

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**If any rule in this document conflicts with a technical constraint, flag it in writing before implementing a workaround. Do not interpret ambiguity as permission to deviate. The platform launches June 2026. Every rule in this document must be implemented and verified before launch.**

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